

A waiting-policy clarification for the recursive auction example

Pathfinder experiment record

28 September 2026

This corrigendum concerns the September 25 repeat–reinjection pilot, reinjection thread Q3P5, ledger entry 19 and its supporting matched-state construction. The archived output is unchanged. This clarification is not an independently reviewed paper or an input to the next experiment.

The gap. After an initial bid of 2, the buyer of value 3 might cross the standing ask of 2.50 on a later invitation. If it does so before the seller of cost 1 arrives, the event that this seller is invited within six draws no longer guarantees a trade with surplus 2. Merely requiring agents to avoid losses does not establish the written lower bound.

An explicit sufficient continuation. Retain the stated eight active traders and their values, the two hidden cost assignments, and the six independent uniform invitations among eligible traders. Fix other traders’ behaviour: they hold their quotes, except that seller S_2 crosses a strictly profitable standing bid when invited. Traders who have transacted exit the round. The acting buyer is allowed to choose its continuation. In state A, an informed comparator posts 2 and leaves it standing for all six draws. Until S_2 is invited there is no trade; all eight traders remain eligible. Consequently its expected added welfare is exactly $2r$, where $r = 1 - (7/8)^6$.

Immediate crossing yields 0.50 in either state and at most another 0.25 in state A, even if later profitable trades are permitted. In state B, after a noncrossing first action no seller can profitably cross a bid below 2.50. A profitable trade requires the buyer to be invited again; its surplus is at most 0.50. Thus that branch has expected welfare at most $0.50r$, while immediate crossing yields 0.50.

Under an equal prior, let p be the probability that an observation-only policy crosses immediately. Its state-informed regret is bounded below by

$$\frac{1}{2} [p(2r - 0.75) + (1 - p)0.50(1 - r)] \geq \frac{1}{2} \min\{2r - 0.75, 0.50(1 - r)\} = 0.1121988296508789 \dots$$

The first term compares with a feasible informed waiting policy, not an assumed optimal policy; the second allows the uninformed buyer to adapt after later observations. This is sufficient for the regret inequality. It does not mean that every noncrossing continuation earns $2r$.

Scope. This repairs the existence construction by specifying behaviour, not by demonstrating that the stipulated behaviour is an equilibrium or typical of the source experiment. The hidden-state prior is illustrative. No frequency estimate, simulator-equivalence test, or causal attribution of observed market losses follows. The original conditional-on-a-final-actor repeat example remains a related, distinct construction.